Big Data Analytics for Energy Usage Assessment in Public School Power Systems: A Case Study of the Western Cape, South Africa

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Abstract. Globally, public school buildings play an important social responsibility. Therefore, understanding and analyzing energy performance in this type of building is essential. So, this study developed a Big Data analytics framework to analyze smart meter data which was collected in 2023 from 53 public schools in the Western Cape, South Africa. The dataset contains 850,000 interval readings, representing a total electricity consumption of 2.51 GWh, together with school metadata. The results show that weekend electricity consumption was relatively high, reaching about 59–65% of weekday usage and it was constant between 01:00 and 04:30. This overnight use accounts for 12–14% of annual electricity use. When it comes to seasonal analysis, the results reveal that the secondary schools had a highest electricity demand during winter from June to August, peaking around midday (11:00–13:00). The analysis also present that the electricity use varied across schools. In addition, a small number of schools used a large share of total electricity demand e.g., one secondary school (S24) showed unusually high power usage overnight compared with similar schools. This highlights a potential way to consider measures to mitigate the energy consumption. Overall, the study demonstrates the importance of scalable big data technologies for analyzing the high-frequency smart meter data and providing useful information for energy management. It also helps identifying inefficiencies, supporting evidence-based decision-making, and improving energy efficiency within public school power systems.

Keywords: Big Data · Smart Metering · Public Schools · MongoDB · PySpark · Tableau · Energy Efficiency · Western Cape · Power Systems · Electricity Consumption Analytics.

1 Introduction

Buildings account for approximately 30% of the world's final energy consumption. Public buildings, including schools, often use a large amount of energy,

but their electricity demand is often not studied or well understood [7]. In South Africa, energy management is difficult because the national electricity grid always faces problems and regular load-shedding events occur [5]. Public schools operate under constrained budgets and therefore require efficient energy management practices [13]. Previous studies have highlighted the importance of energy-efficiency initiatives in schools and public buildings [1], [4], [11], while emerging digital technologies are increasingly being used to support data-driven decision-making in educational environments [9].

Despite these advances, most building-energy studies focus on residential, commercial, or university settings and often rely on low-resolution billing data [6], [2], [12]. Although smart meters generate large volumes of high-frequency data, few studies have applied scalable big data technologies to analyze electricity consumption in public schools [10]. Consequently, there remains a lack of integrated frameworks that combine high-resolution smart-meter data, distributed processing, and interactive visualization for school energy management. To address this gap, this study develops an end-to-end big data analytics framework based on MongoDB [3], PySpark [8], and Tableau [14] and applies it to half-hourly electricity-consumption data from 53 public schools in the Western Cape. The framework is used to analyze consumption patterns, load profiles, seasonal variation, peak demand, baseload consumption, and energy-efficiency interventions to support evidence-based energy management in public school power systems.

The remainder of this paper is organized as follows. Section 2 describes the research methodology adopted in this study. Section 3 presents and discusses the results of the analysis. Finally, Section 4 concludes the paper and outlines directions for future research.

2 Research Methodology

This section describes the research methodology employed in this study and outlines the procedures followed for data collection, storage, preprocessing, analysis, and visualization to obtain the reported analysis.

2.1 Proposed Big Data Analytics Architecture

To analyze electricity consumption in public schools in the Western Cape, South Africa, a generic big data analytics architecture was proposed, as illustrated in Figure 1. The architecture consists of seven stages: (i) data collection (acquisition), (ii) Data ingestion, (iii) data preprocessing, (iv) curated storage, (v) analytics, (vi) aggregation, and (vii) visualization. The details of each of these stages are described the following subsections.

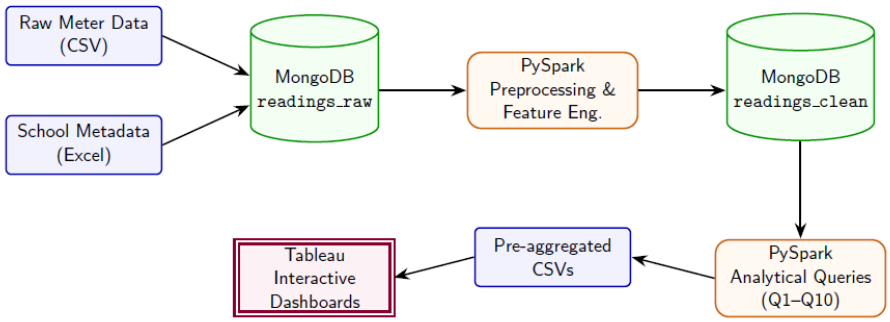


Fig. 1: Proposed Big Data Analytics Framework for Energy Usage Analysis in Public Schools.

2.2 Dataset Description

The dataset used in this study was released by Stellenbosch University in 2024 and contains half-hourly electricity consumption readings from 53 public schools in the Western Cape, South Africa and can be found here[3].

Table 1: Summary of the school-level metadata and 2023 readings.

Attribute	**Value(s)**	**Coverage**
Number of schools	53	100%
Quintile distribution	Q1=18, Q2=15, Q3=17, Q4=3, Q5=0	53 / 53
School type	primary, secondary, combined, intermediate	53 / 53
Learners per school	356–1894 (mean 1009)	53 / 53
Educators per school	8–65 (mean 30)	53 / 53
Intervention recorded	yes = 43, no = 10	53 / 53
Reading interval	30 minutes	—
Rows in 2023	849,696	all schools
Total energy 2023	2,505,295 kWh ≈ 2.51 GWh	all schools
Mean annual kWh / school	≈ 47,270 kWh	—

The data spans the period from December 2022 to early 2024 and is organized into three components: *raw data*, *clean data*, and *enhanced data*. The raw files contain the original meter readings and missing values, the clean files include publisher-supplied imputations for short outages, and the enhanced files provide additional attributes such as month, season, school term, school-day status, and holiday classification. School-level characteristics are provided in a separate `metadata.xlsx` file.

[3] https://data.mendeley.com/datasets/nm2vc4v2br/1

The dataset covers 53 schools distributed across four funding quintiles, with 18, 15, 17, and 3 schools belonging to Quintiles 1, 2, 3, and 4, respectively. The participating institutions comprise primary, secondary, combined, and intermediate schools. Learner enrollment ranges from 356 to 1,894 learners per school. In addition, 43 of the 53 schools are recorded as having received at least one energy-efficiency intervention. A summary of the school metadata is presented in Table 1. The dataset has two notable limitations. First, no Quintile 5 schools are included in the sample, restricting the generalizability of the findings to Quintiles 1-4. Second, Quintile 4 is underrepresented, with only three schools available for analysis.

2.3 Data Processing

Following dataset curation, a series of preprocessing techniques were applied to improve data quality and prepare data for subsequent analysis. These techniques are described in the following subsections.

Missing Values The dataset contained a limited number of missing readings resulting from temporary meter outages and communication failures. The original records were retained unchanged in MongoDB, while data cleaning and imputation were performed within the PySpark processing pipeline. Missing values were imputed using the "median" electricity consumption for the corresponding school. The median was selected as the imputation measure because of its robustness to skewed distributions and extreme observations that are common in electricity consumption data. Unlike the mean, the median is less affected by occasional consumption spikes and therefore provides a more reliable estimate of the typical energy-use level. Furthermore, school electricity demand exhibits clear temporal variations between weekdays, weekends and holiday periods. Therefore, median-based imputation helps preserve the overall consumption pattern without introducing unrealistic values or altering the underlying data distribution.

Handling Outliers After treating the missing values, the outliers in the dataset were also handled. During the examination of the data, a small proportion of observations showed unusually high electricity-consumption values compared with the normal range of the school. So, these values were considered possible meter or data transmission errors. To reduce the effect of these extreme values, a percentile-capping technique was used. Values above the 99.9th percentile of each school's annual electricity use were replaced with the 99.9th percentile value. This approach helped reducing the effect of extreme outliers while keeping genuine high-demand events in the data. Percentile capping is particularly suitable for energy-consumption datasets, where genuine high-demand period may occur and should not be removed, but measurement errors can distort statistical analyses and machine-learning models. Consequently, the adopted approach improved data quality while maintaining the overall characteristics of the consumption patterns.

Handling Inconsistent Data After handling the outliers, the dataset was checked for errors in the derived attributes. During this process, some inconsistencies were found in the enhanced dataset. For example, records for School S46 had duplicated values in the `season` and `schoolDay` fields such as `Autumn` and `yes`. To fix these errors, the `season` and `schoolDay` fields were recreated from the timestamp using PySpark.

Normalization of Data Types and Timestamps After fixing the data quality issues, the data types and timestamp values were standardized. In the dataset, the timestamp values are originally stored as strings in the format `YYYY-MM-DD HH:MM:SS`. These values were converted and set to the *Africa/Johannesburg* time zone to match the local time of the participating schools. In addition, the electricity consumption attribute (`kWh`) was converted to the `double` data type, while categorical variables such as "school quintile" and "intervention status" were kept as string variables throughout the analysis.

Feature Engineering After standardization, additional features were created to support the analysis of electricity consumption. The timestamp attribute was used to create several time-related features, including year, month, day, hour, and day of the week. A binary `is_weekend` variable was also created to show whether a reading occurred on a weekend. Each record was assigned to a Southern Hemisphere meteorological season: Summer (December-February), Autumn (March-May), Winter (June-August), or Spring (September-November). To support school-calendar-based analyses, the data were also mapped with the 2023 Western Cape Department of Education academic calendar and each record was classified as either *Term 1-4* or *Holiday*. In addition, a binary `schoolDay` variable was created to show whether the reading occurred on a weekday during an official school term.

3 Results and Discussion

In this section, the results obtained from the analysis of electricity-consumption are presented and discussed.

3.1 Annual Electricity Consumption and Quintile Differences

Table 2 summarizes the analysis results obtained to understand on the annual electricity consumption by funding quintile. Quintile 1 schools account for the largest share of total electricity consumption, with a combined annual usage of 874.1 MWh, followed by Quintile 3 schools (815.1 MWh) and Quintile 2 schools (688.3 MWh). However, at the individual school level, mean annual consumption is relatively similar across the quintiles, ranging from 42,610 kWh in Quintile 4 to 48,561 kWh in Quintile 1. Median annual consumption also showed a similar trend, indicating that funding classification alone does not strongly influence

school-level electricity demand. While these observations suggest comparable consumption patterns across quintiles, no formal statistical significance test was performed. Therefore, they should be interpreted as descriptive trends rather than evidence of statistically significant differences and may not be fully generalizable beyond the schools included in this study.

Table 2: Annual Electricity consumption by School Funding Quintile (2023).

Quintile	**n**	**Total MWh**	**Mean kWh**	**Median kWh**
Q1	18	874.1	48,561	39,783
Q2	15	688.3	45,885	42,180
Q3	17	815.1	47,947	40,291
Q4	3	127.8	42,610	39,374

As shown in Figure 2, the mean and median electricity consumption vary only marginally across funding quintiles. This observation suggests that factors such as operational requirements, facility characteristics, learner enrollment, and school type may have a greater influence on electricity demand than socioeconomic classification. Consequently, funding quintile alone does not appear to be a strong predictor of school-level energy consumption. However, comparisons involving Quintile 4 should be interpreted with caution due to the relatively small number of schools represented in this category.

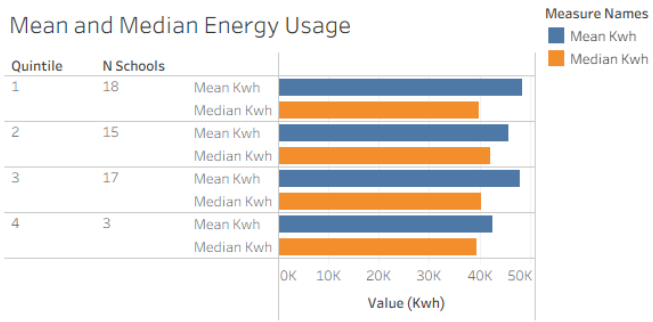


Fig. 2: Mean and Median Annual Energy Consumption by Funding Quintile.

Although the funding quintiles show socioeconomic backgrounds of schools, electricity consumption can also vary by type of the school (intermediate, primary, and secondary) and also based on number of students in that school. There-

fore, it is important to examine and compare electricity consumption across these school types and energy use per-student. The results are shown in Figure 3 and from this, it can be observed that secondary schools have the highest average annual electricity consumption per school (about 61,600 kWh, compared with about 41,800 kWh for primary schools). This may be due to their greater operational needs such as the use of computer and science laboratories, specialized equipment, and may be larger campus footprints. However, total electricity consumption alone does not necessarily reflect energy performance, because schools have different number of students. Therefore, electricity consumption should also be considered per student to provide a better measure of energy efficiency.

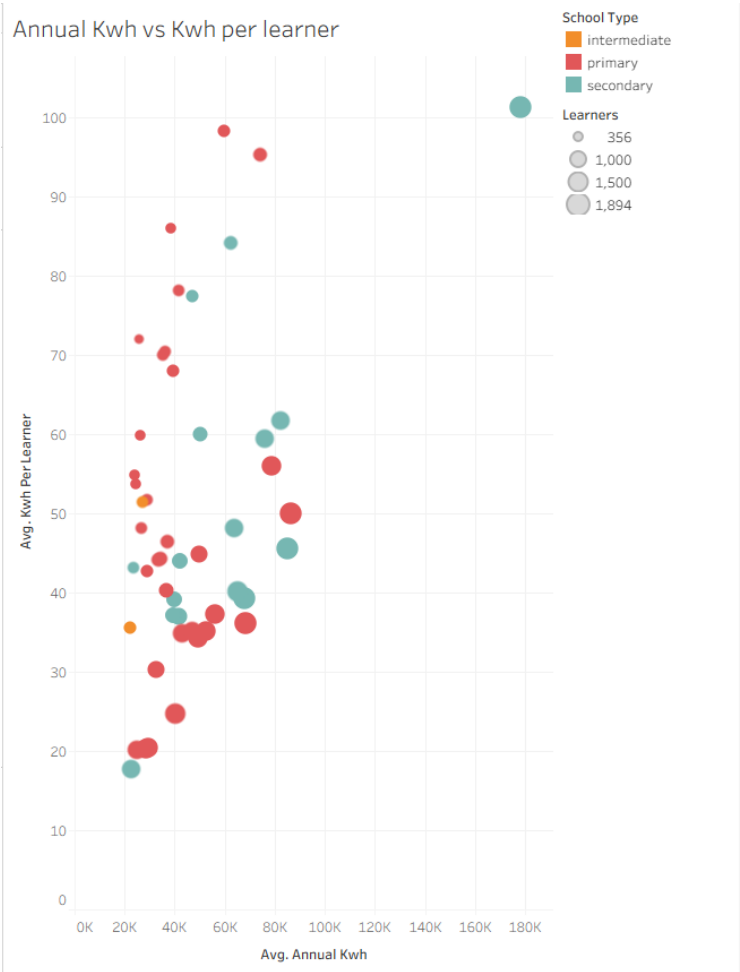
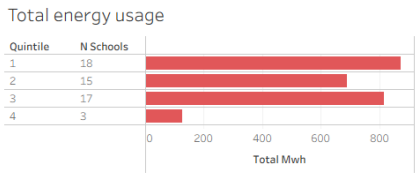


Fig. 3: Annual electricity consumption and efficiency by school type and enrollment.

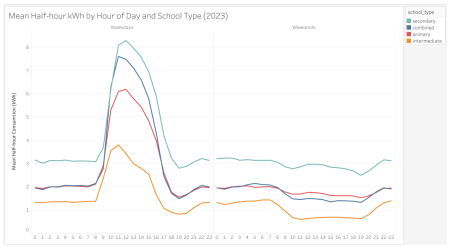
Total electricity consumption shows the overall electricity demand of a school, but it does not account for differences in school size. Therefore, Figure 3 also shows the electricity consumption per-student. The results show that schools with similar total electricity consumption can have different levels of energy use per student. As a result, the per-learner electricity consumption provides a better measure of energy use and can help identify schools that may need further investigation or energy efficiency measures.

3.2 Daily and Weekly Load Profile

The daily and weekly load-shape results are shown in Figures 4a and 4b. From these figures, it is observed that the average daily load profile shows a clear school-day pattern. Electricity demand remains relatively stable overnight (00:00–06:00), increases sharply between 09:00 and 11:00, and remains relatively high during teaching hours. After about 14:00, electricity use gradually decreases as fewer people may remain at the school. Across all school categories, weekday peak demand occurs between 11:00 and 12:00. Secondary schools exhibit the highest average peak demand (8.29 kWh per 30-minute interval, at 12:00), followed by combined schools (7.61 kWh, at 11:00), primary schools (6.19 kWh, at 12:00) and intermediate schools (3.80 kWh, at 11:00). These differences likely reflect variations in school size, infrastructure, and operational intensity.



(a) Total energy usage by quintile.



(b) Mean half-hourly kWh by hour of day across school types.

Fig. 4: Electricity consumption patterns by funding quintile and school type.

Table 3 shows that weekend consumption is substantially lower than weekday demand, but still averages 59–65% of weekday demand across quintiles. The relatively flatter weekend profiles and persistent background consumption suggest that occupancy and school activities are important drivers of demand, while a substantial baseline load remains outside school hours. As shown in Table 3, weekend mean demand remains between 59% and 65% of the weekday mean across all quintiles. This indicates that more than half of typical weekday consumption persists during weekends, when classes are not in session. A portion of this residual demand may be associated with continuous loads such as security

lighting, networking equipment, refrigeration, and idle HVAC systems, making these loads potential targets for low-cost interventions such as circuit-level timers and improved scheduling.

Table 3: Weekend versus weekday mean half-hour demand by quintile (2023). The weekend ratio is the mean of the per-school weekend/weekday ratios.

Quintile	n	Avg. Weekday (kWh/30 min)	Avg. Weekend (kWh/30 min)	Weekend Ratio
Q1	18	3.417	2.205	0.647
Q2	15	3.345	2.011	0.604
Q3	17	3.409	2.162	0.585
Q4	3	3.097	1.817	0.614

3.3 Peak Demand and Baseload

Figure 5 presents the highest half-hour electricity-demand observations recorded during 2023. Peak electricity demand events were primarily observed in secondary schools, particularly Schools S43, S48, and S47, and occurred predominantly in winter, between 10:00 and 14:00 (mostly around midday).

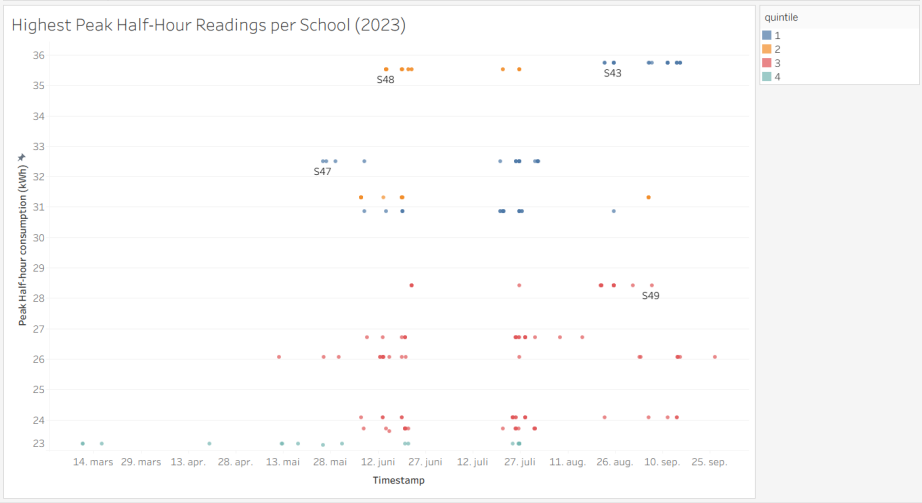


Fig. 5: Peak demand and baseload characteristics across schools.

The highest half-hourly consumption in the cleaned data was approximately 36 kWh (School S43), corresponding to an average power demand of approximately 71 kW. These elevated winter peaks are likely attributable to increased

heating and lighting requirements associated with lower temperatures and recorded daylight hours. Overnight consumption during 01:00 - 04:30 accounts for approximately 12 - 14% of annual energy use, indicating a substantial and persistent baseload. Similar levels across quintiles suggest that these background loads are common across schools. School S24 is a notable exception, with an average overnight demand of approximately 9.3 kWh per 30-minute interval (about 19 kW) and the highest annual electricity consumption. This persistent load makes S24 a strong candidate for a detailed energy audit, with potential savings from identifying and reducing unnecessary overnight consumption.

3.4 Top Consumers and Interventions

Figure 6 further shows the dominance of secondary schools among the highest consumers, suggesting that targeted interventions at a small number of high-consuming schools could deliver substantial system-level savings.

Table 4 shows that the ten highest-consuming schools account for approximately 34% of total electricity use across the 53 schools, indicating a highly concentrated consumption pattern. School S24 is the largest consumer, using 178,043 kWh in 2023, more than twice the consumption of the second-ranked school despite having a similar learner population. This is consistent with its unusually high overnight baseload.

Table 4: Top 10 annual consumers (2023).

Rank	School	Type	Quintile	Learners	Annual kWh	kWh/Learner
1	S24	Secondary	Q3	1,758	178,043.3	101.28
2	S49	Primary	Q3	1,728	86,492.76	50.05
3	S43	Secondary	Q1	1,861	84,844.99	45.59
4	S48	Secondary	Q2	1,334	82,253.8	61.66
5	S3	Primary	Q1	1,405	78,651.81	55.98
6	S47	Secondary	Q1	1,276	75,844.86	59.44
7	S41	Primary	Q1	778	74,094.26	95.24
8	S25	Primary	Q2	1,894	68,502.18	36.17
9	S17	Secondary	Q2	1,730	68,042.57	39.33
10	S14	Secondary	Q4	1,619	64,938.76	40.11

Table 5 shows that schools with recorded interventions consumed approximately 4,500 kWh less annually on average than those without interventions (46,424 versus 50,906 kWh). Lower consumption among intervened schools is particularly evident in Q1 and Q2. In Q3, however, intervened schools have higher average consumption, largely due to S24. Excluding S24 reduces the Q3 intervened average from 49,681 to approximately 40,500 kWh and brings it closer to the pattern observed in the other quintiles.

These results suggest a potential association between interventions and lower electricity consumption, but do not establish causality. The absence of informa-

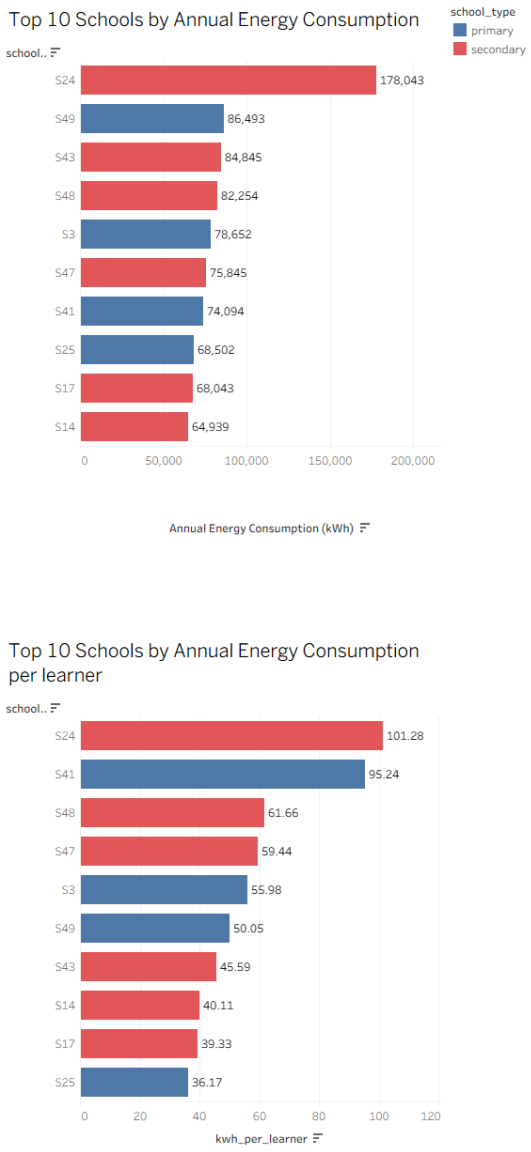


Fig. 6: Bar chart of the top 10 annual energy consumers.

tion on intervention type, timing, and pre-intervention consumption limits direct assessment of intervention effectiveness.

Table 5: Mean annual electricity consumption by intervention status, overall and per quintile (2023).

Quintile	**Intervention = No**		**Intervention = Yes**	
	Mean kWh	**n**	**Mean kWh**	**n**
Q1	56,673	4	46,243	14
Q2	57,706	3	42,930	12
Q3	34,939	2	49,681	15
Q4	39,374	1	44,228	2
All	50,906	10	46,424	43

4 Conclusions and Future Work

In this study, a big data analytics architecture was developed for analyzing electricity consumption in public schools across the Western Cape. The architecture used MongoDB, PySpark, and Tableau to process approximately 850,000 half-hourly electricity records collected from 53 schools. The results revealed clear differences in electricity consumption across schools and seasons. Weekend consumption was about 59-65% of weekday consumption, while overnight consumption account for about 12-14% of annual electricity consumption. In addition, electricity consumption was highest during winter, particularly in secondary schools around midday. A small number of schools also accounted for a large share of total electricity use. The ten highest consuming schools had approximately 34% of overall consumption, while School S24 showed particularly high electricity use and overnight consumption. Schools with energy-efficiency measures generally had lower electricity consumption. However, limited information about these measures made it difficult to determine their direct impact. The study was also limited by the small number of Quintile 4 schools, the absence of Quintile 5 schools, and the lack of building floor-area data.

To extend this work, the following areas are suggested as future work. First, additional information on building characteristics, occupancy, and infrastructure should be included. Second, a more detailed information about energy-efficiency measures should be collected to evaluate their impact. Finally, the framework could be extended to streaming technologies such as Apache Kafka and Spark Structured Streaming to support near real-time monitoring and detection of unusual electricity consumption.

Acknowledgments

ChatGPT (OpenAI)[4]. was used during the preparation of this manuscript to improve grammar, clarity, and sentence structure. All technical content, analysis, and conclusions are our own work. The suggestions from ChatGPT were carefully reviewed and used only where appropriate, without altering the technical meaning of the paper.

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